

Why is Health Hard for AI?

Unique challenges applying large models to real world wellness

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*Dr. Smarr has a Col with Ouraring Inc.

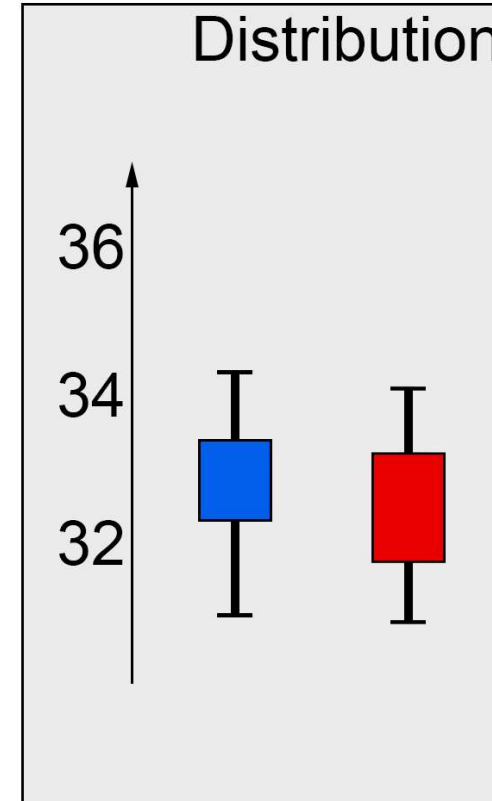
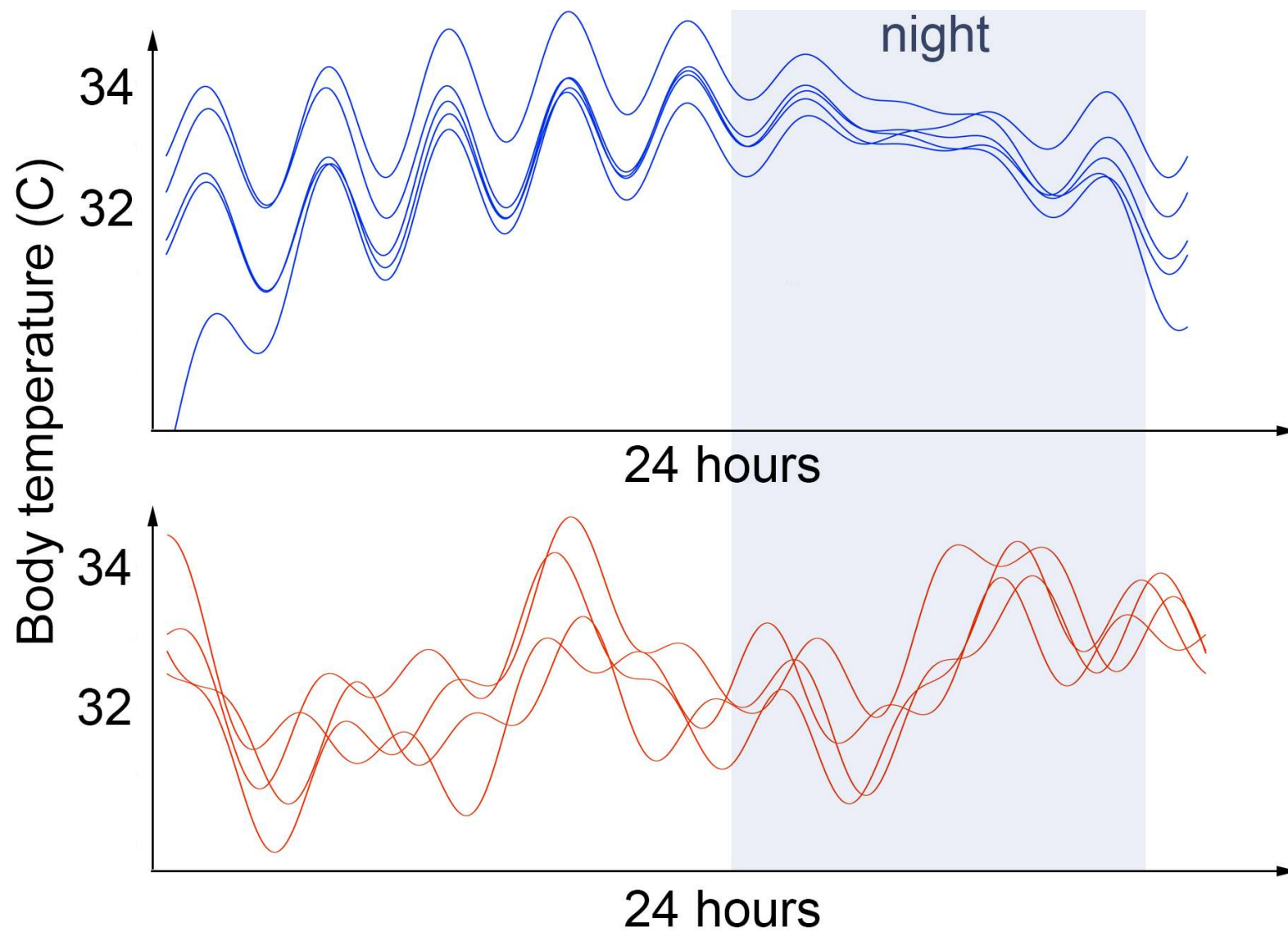


Center for Healthy Aging & Stein Institute for Research in Aging

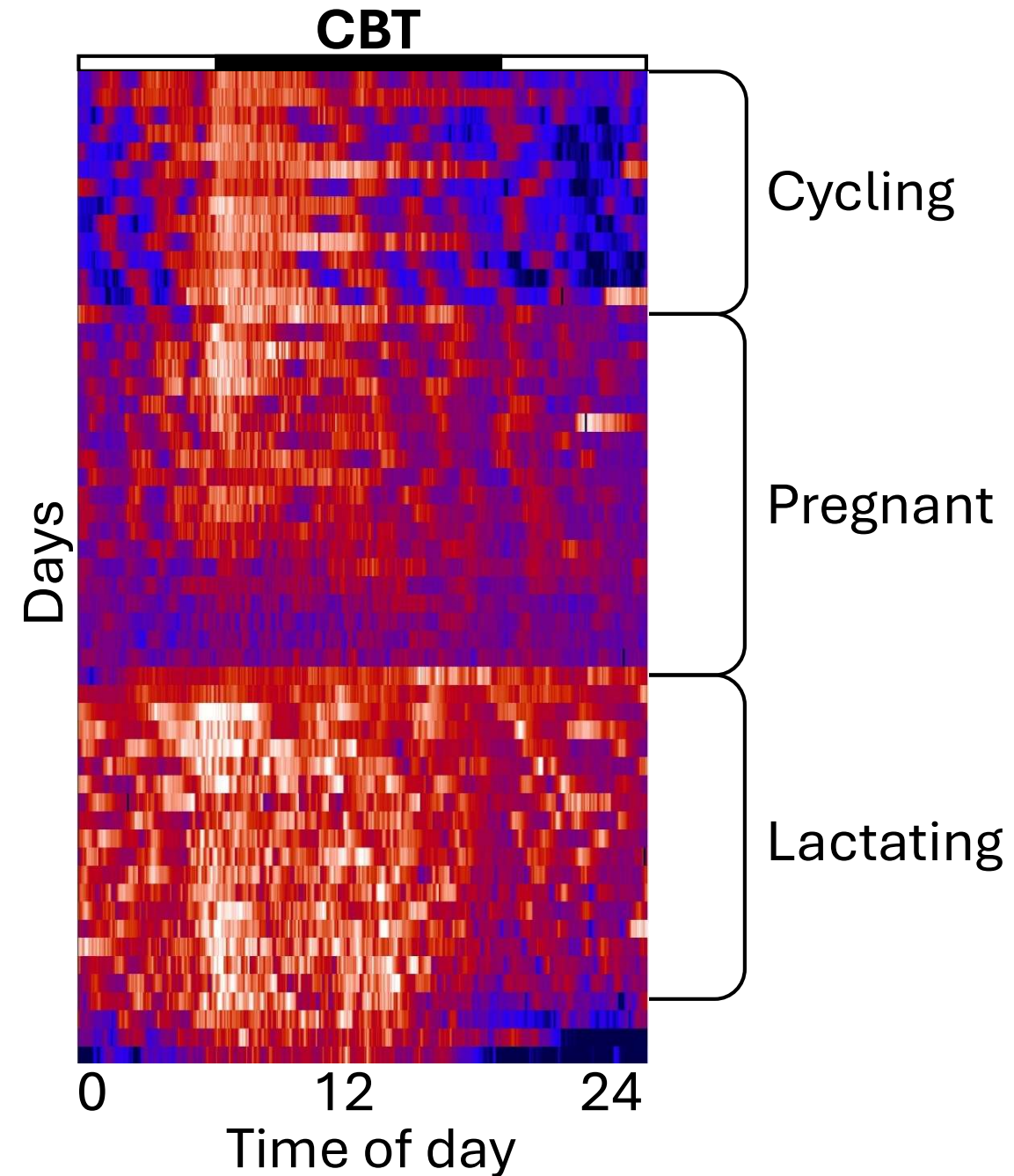
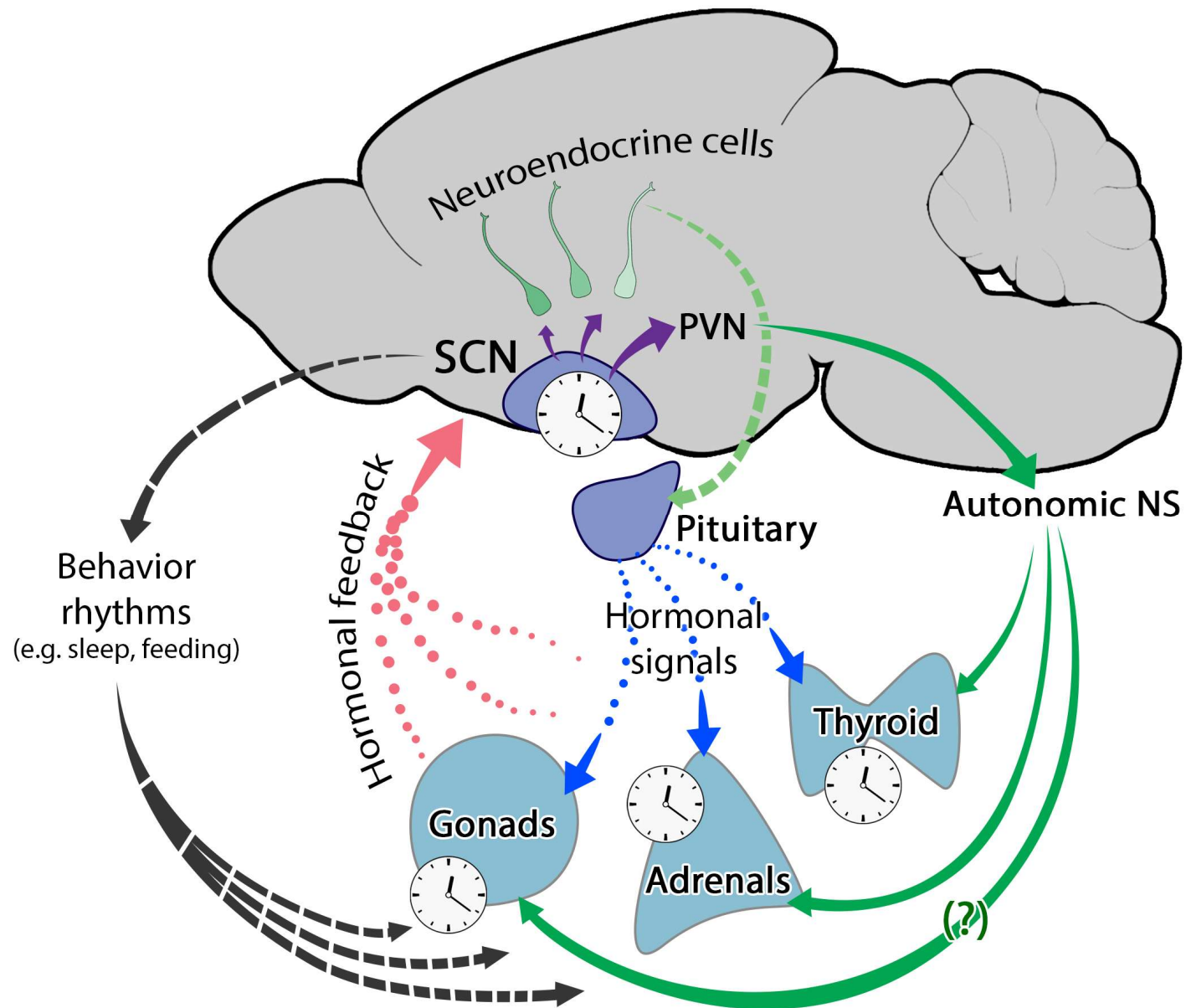
SDSC



New tools let us see new patterns

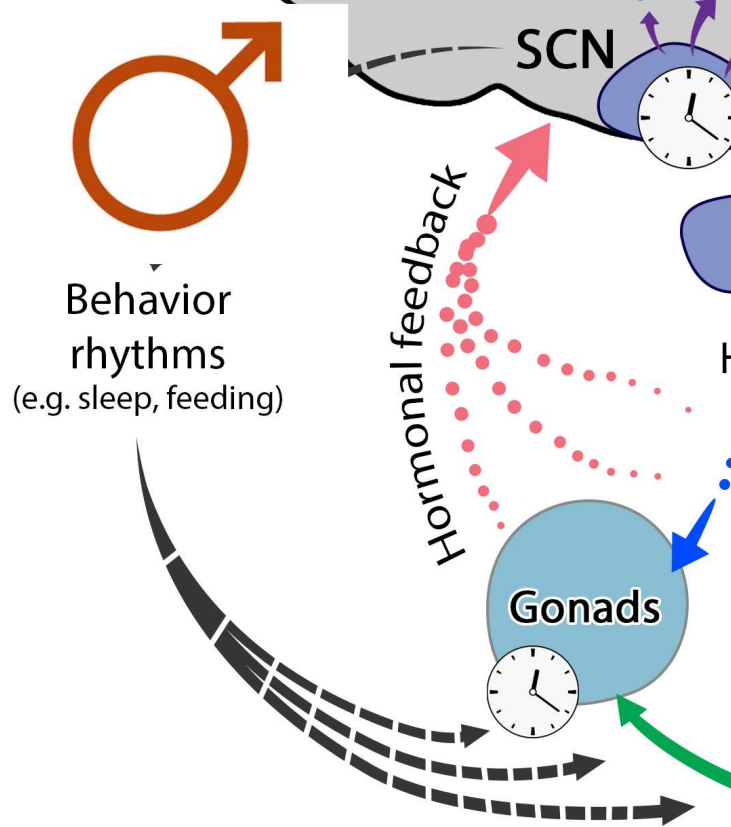


Your body is not random or still



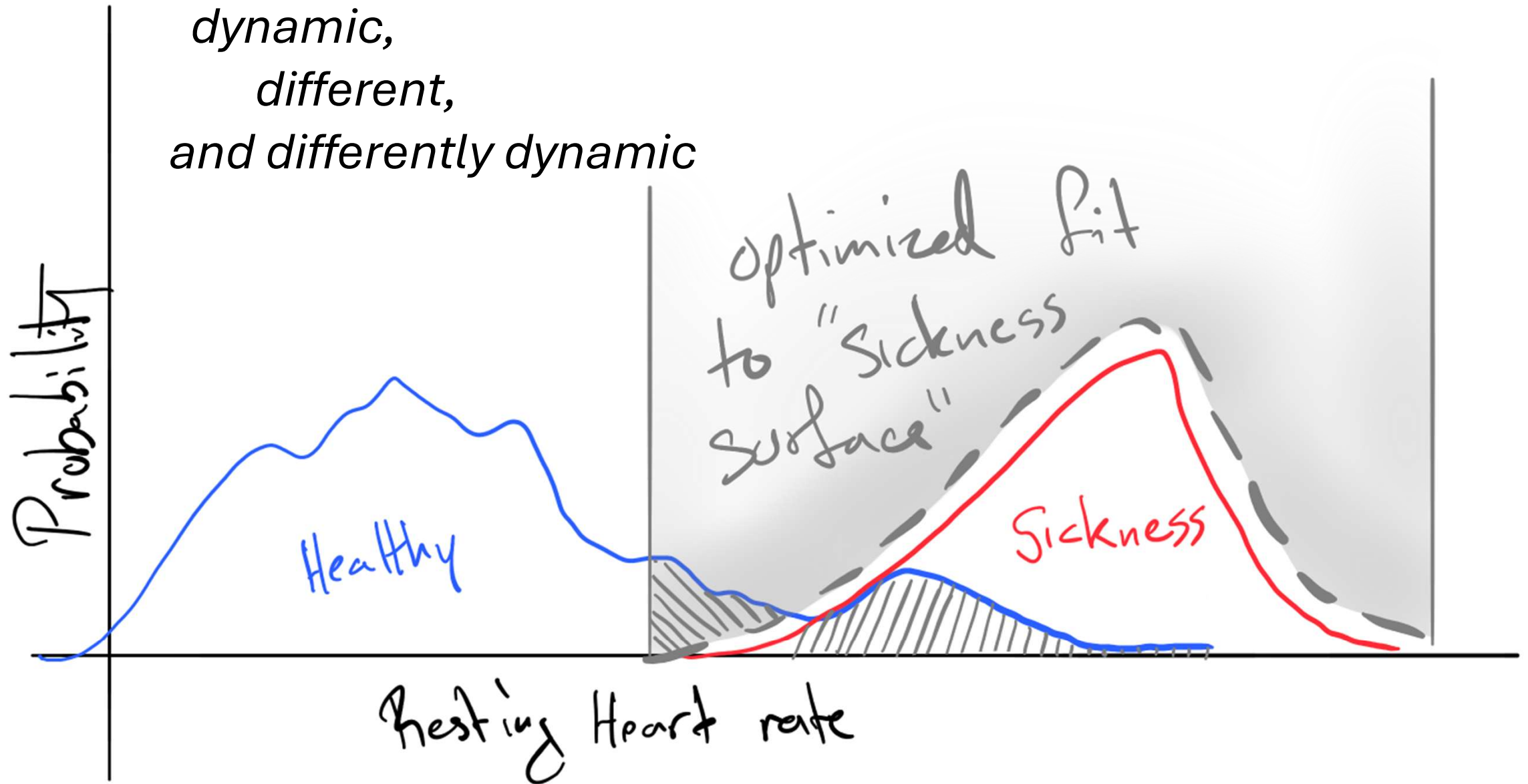
Your body is not random or still

CBT



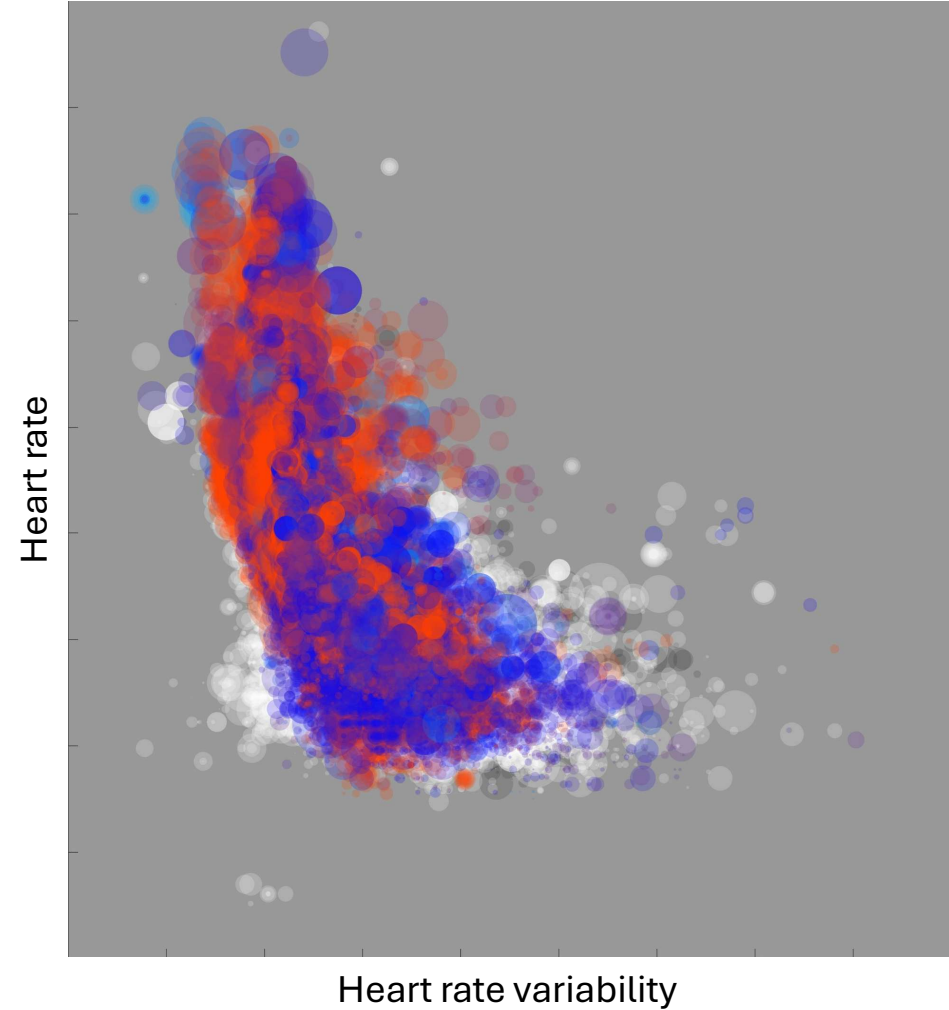
How do algorithms see you: *Feature Statistics*

*Individuals' surfaces are
dynamic,
different,
and differently dynamic*

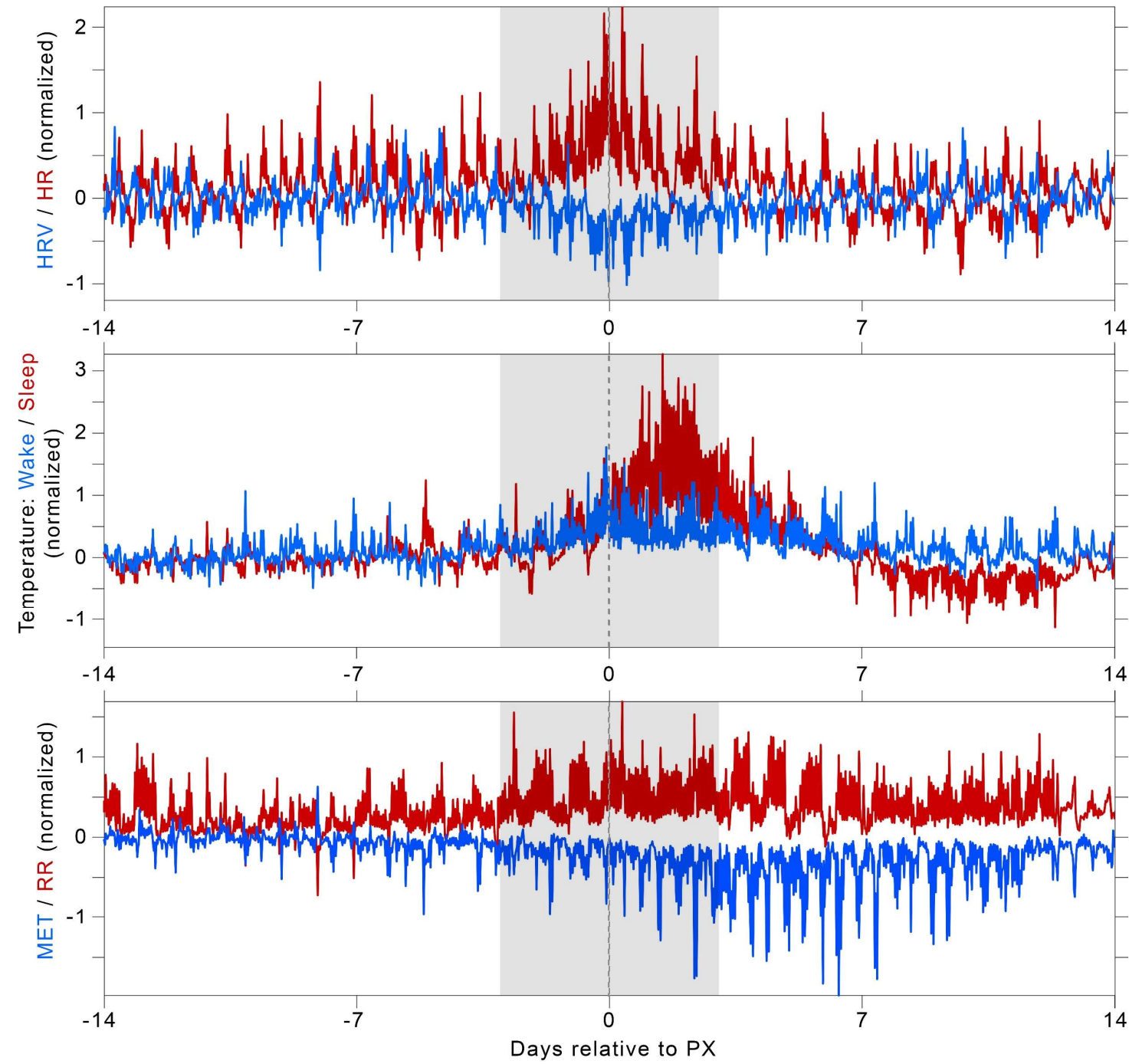


COVID detection

“data landscape”

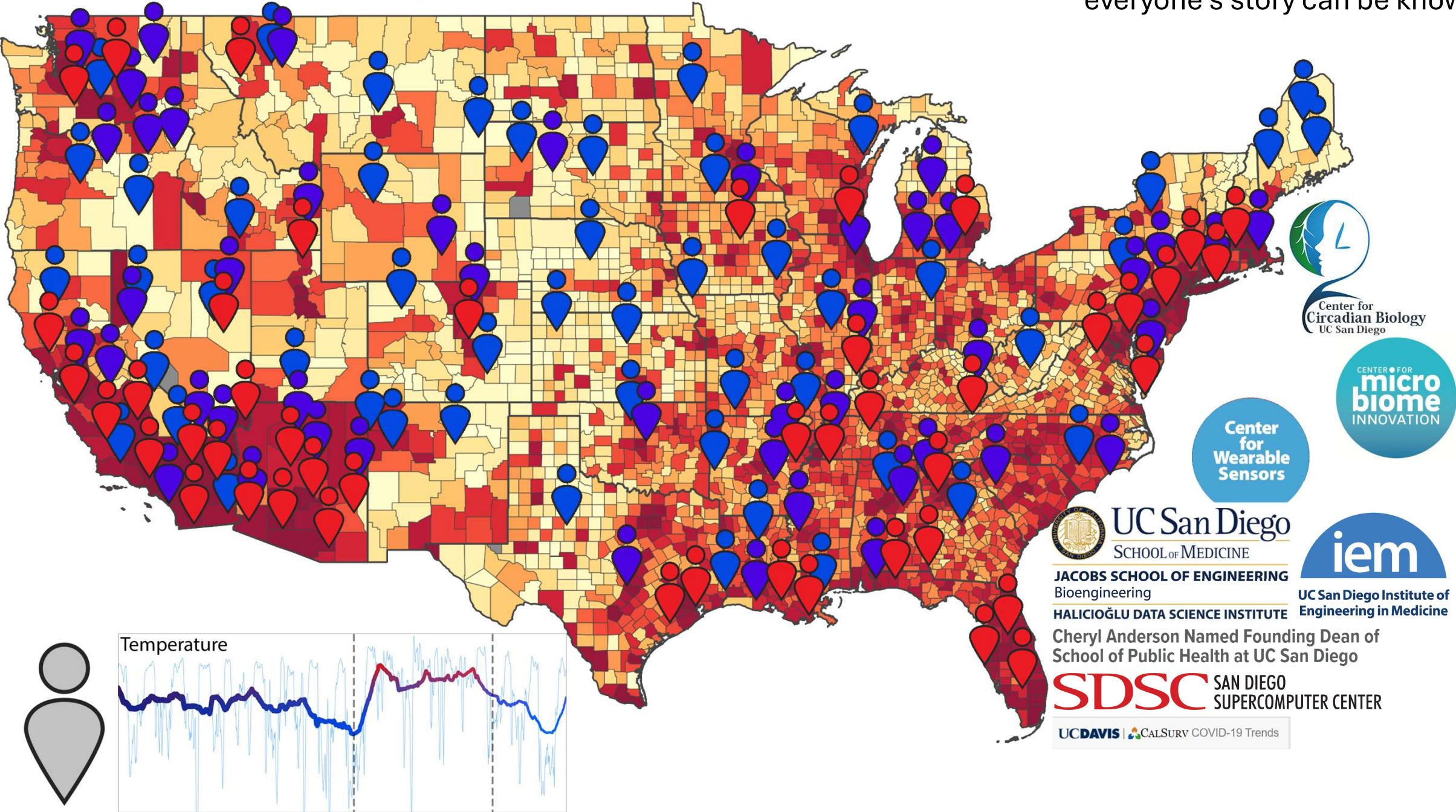


Multimodal wearables data

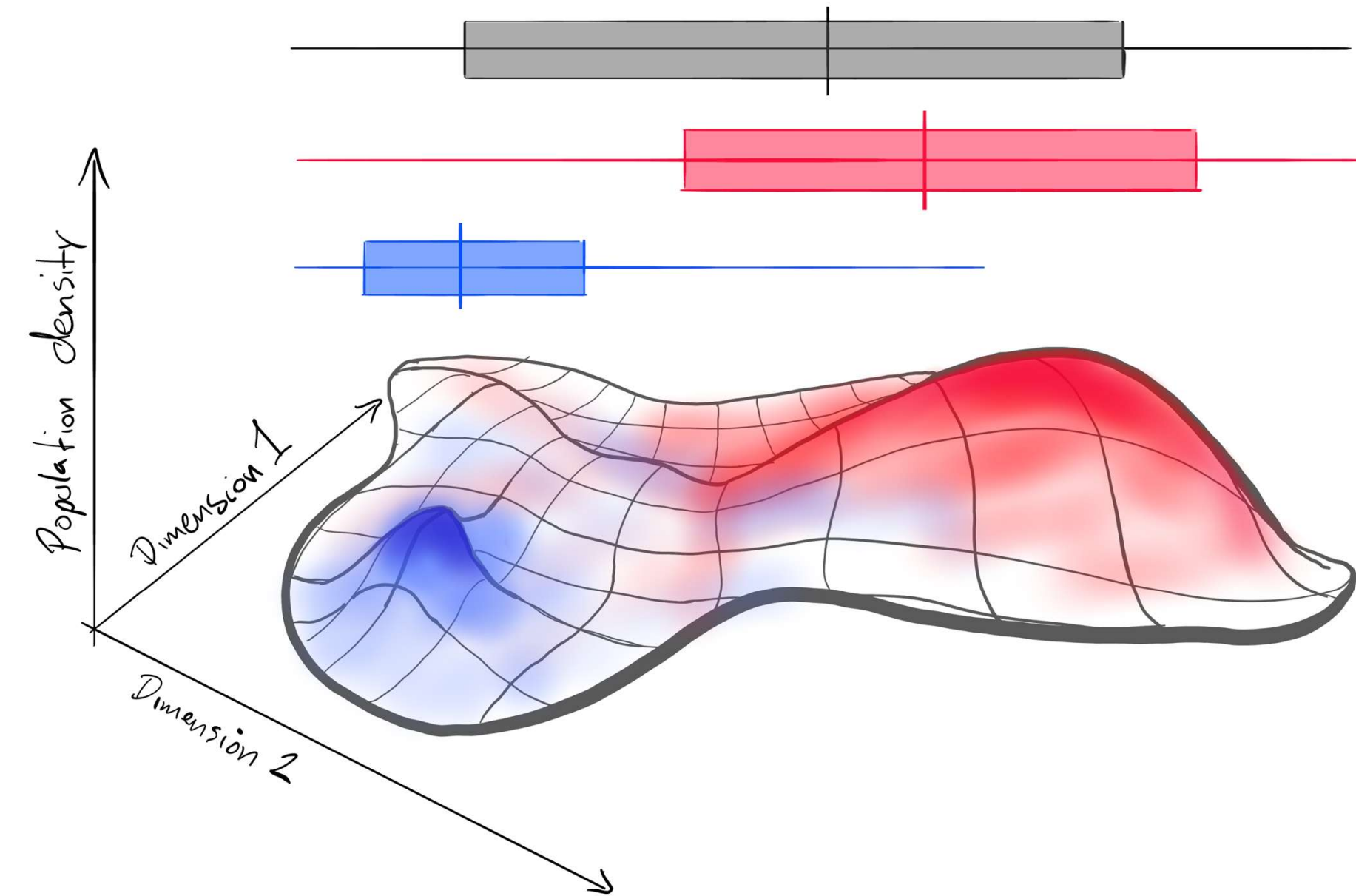


Public health, down to the individual.

Working with communities to ensure
safe, equal coverage, and enable
everyone's story can be known.



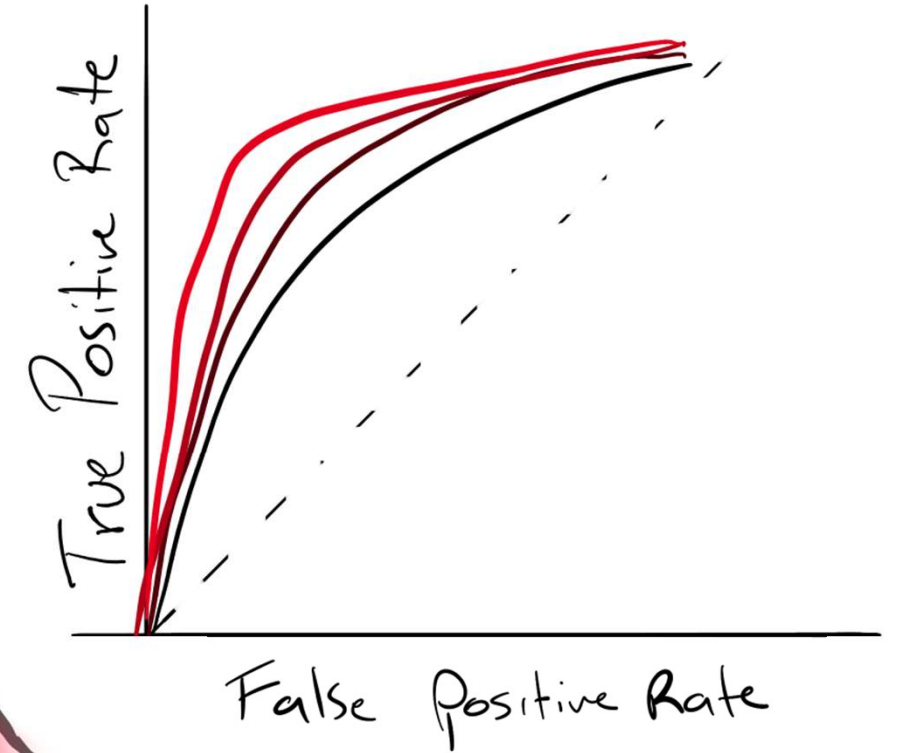
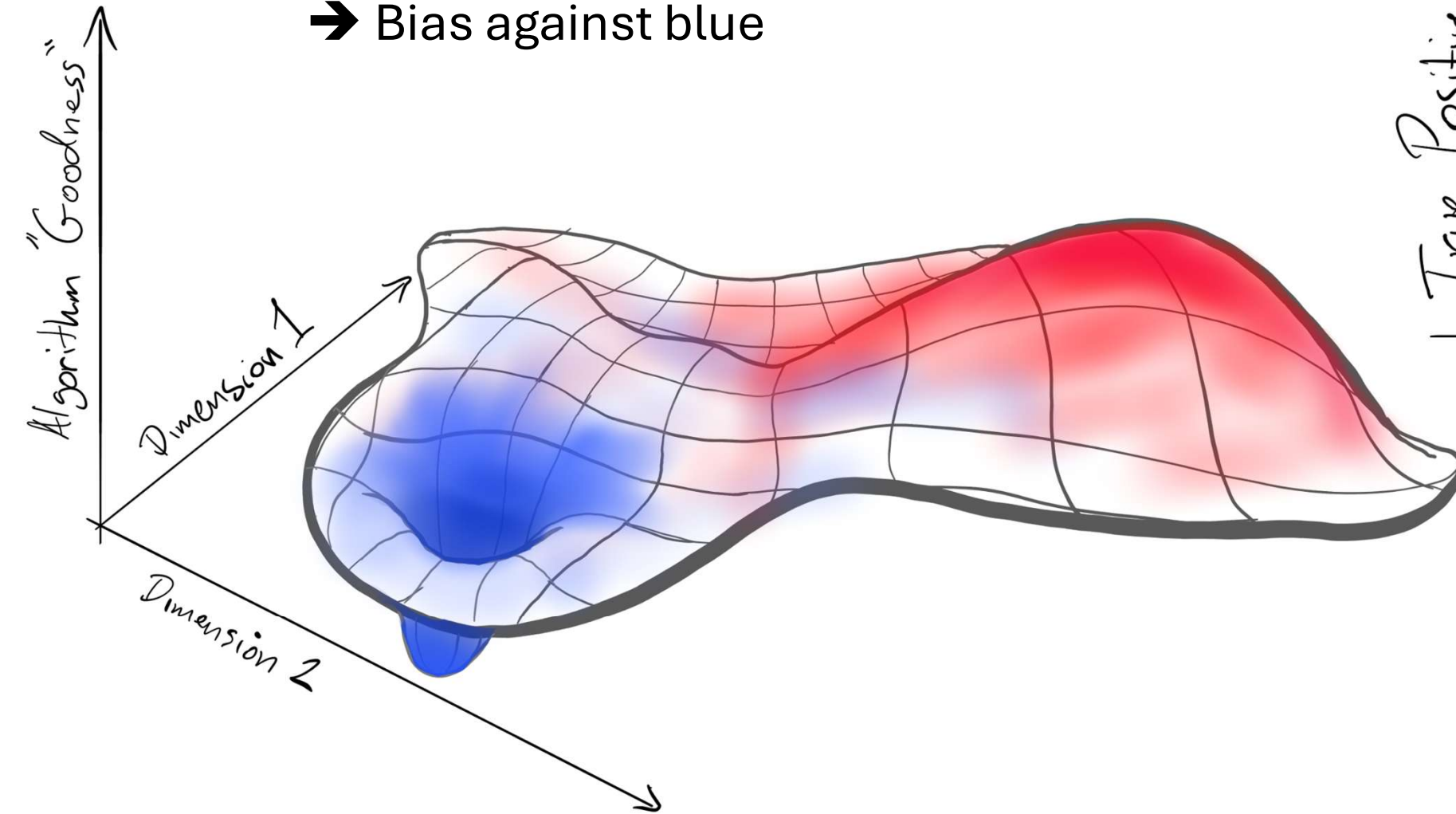
Seeing what the AI sees: numerical landscapes



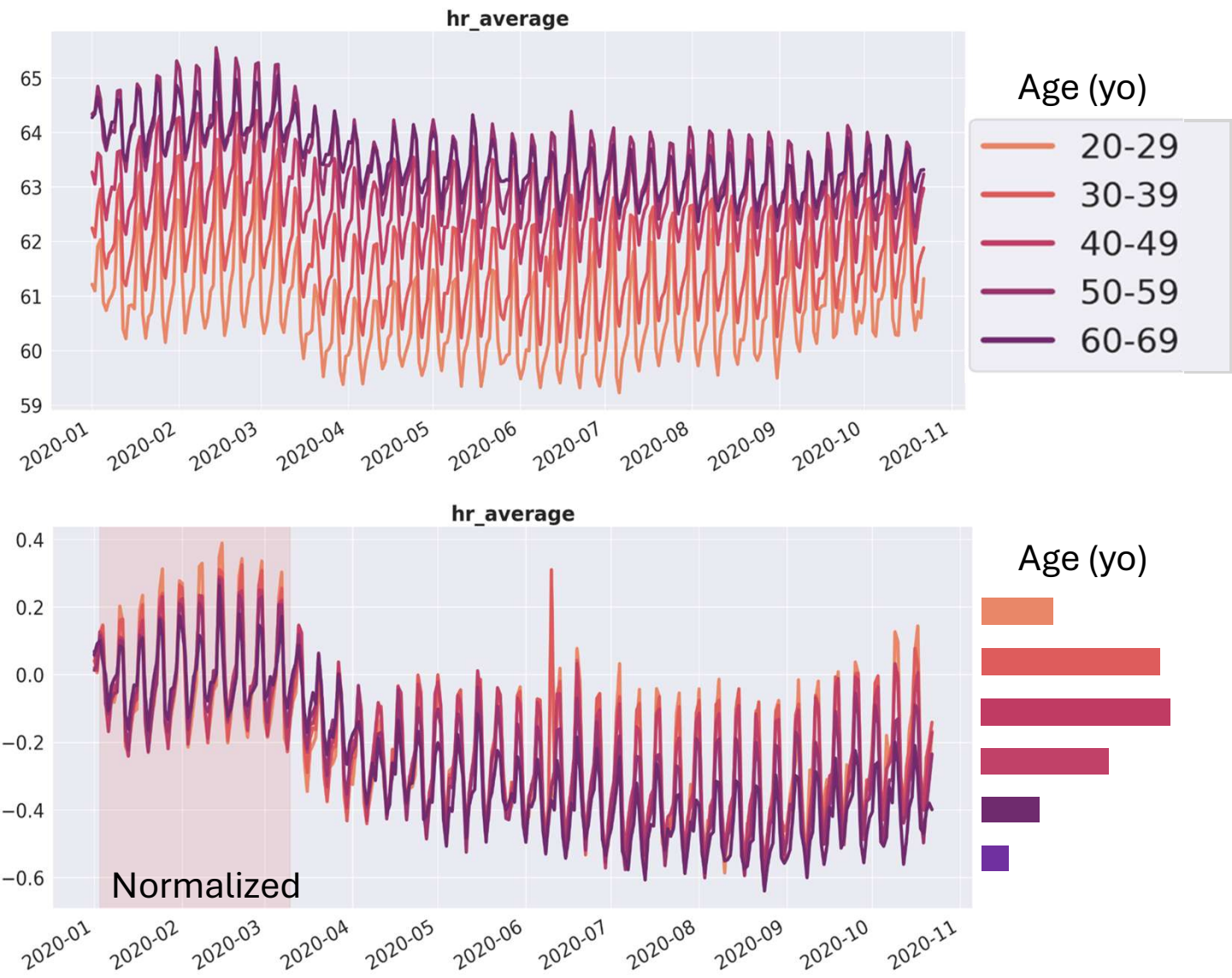
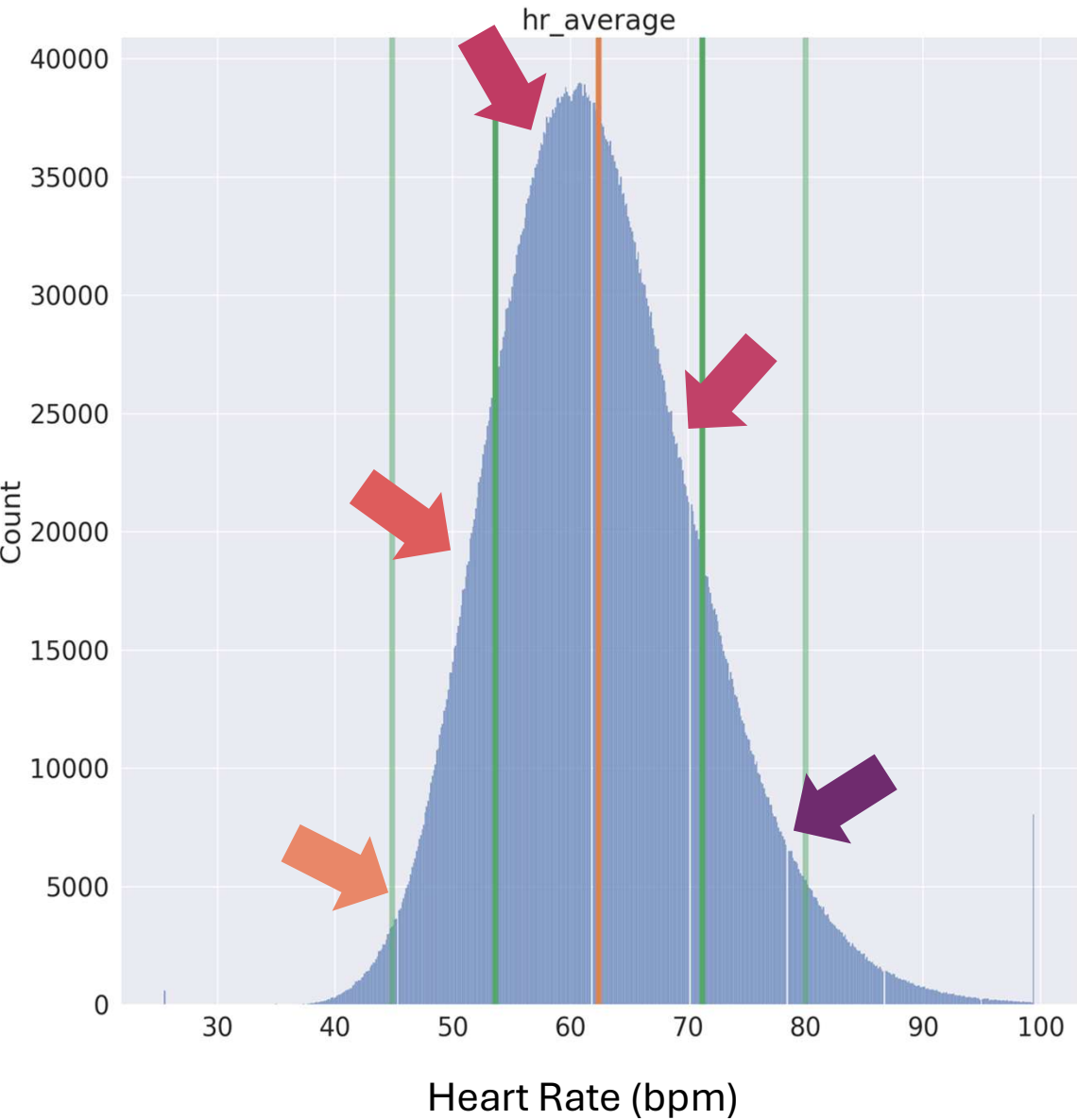
A naïve algorithm will look for the *global* maximum

Red population is closer to overall mean,
so betting on red is a winning bet

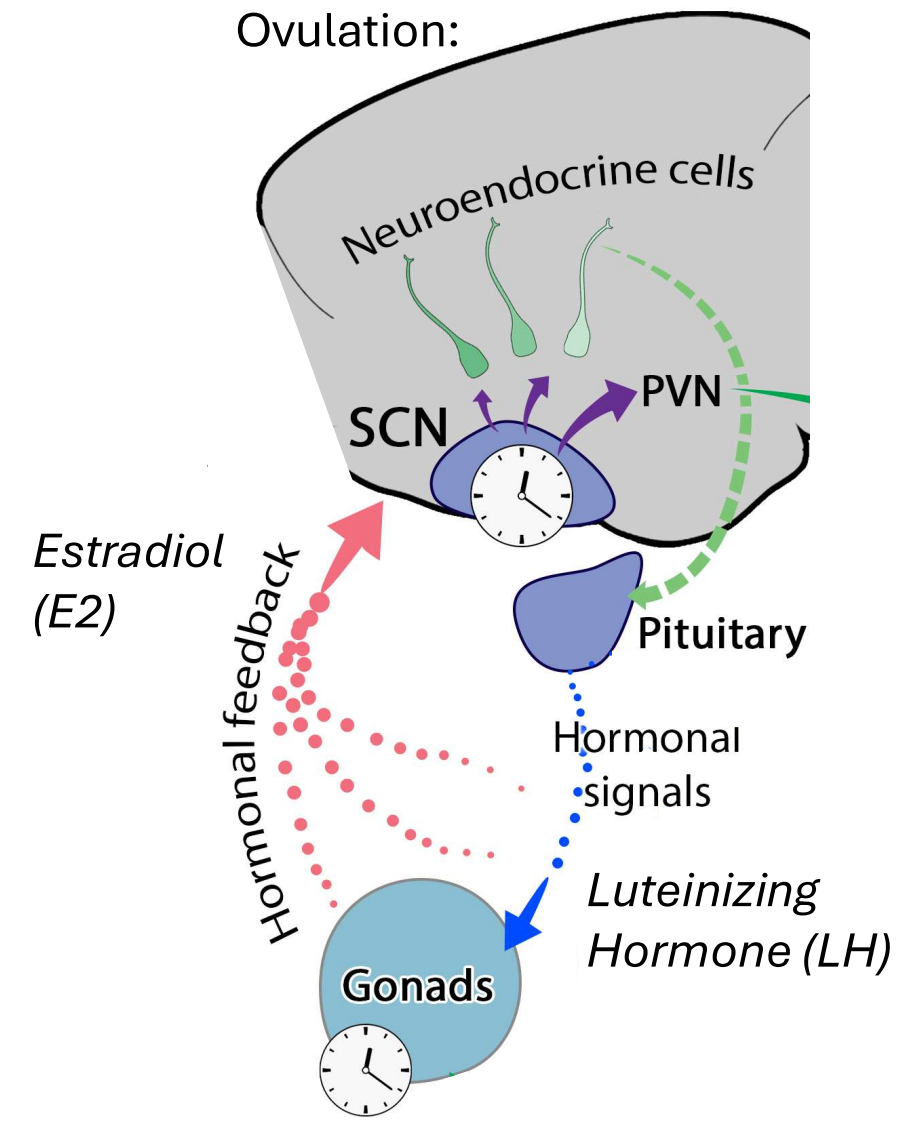
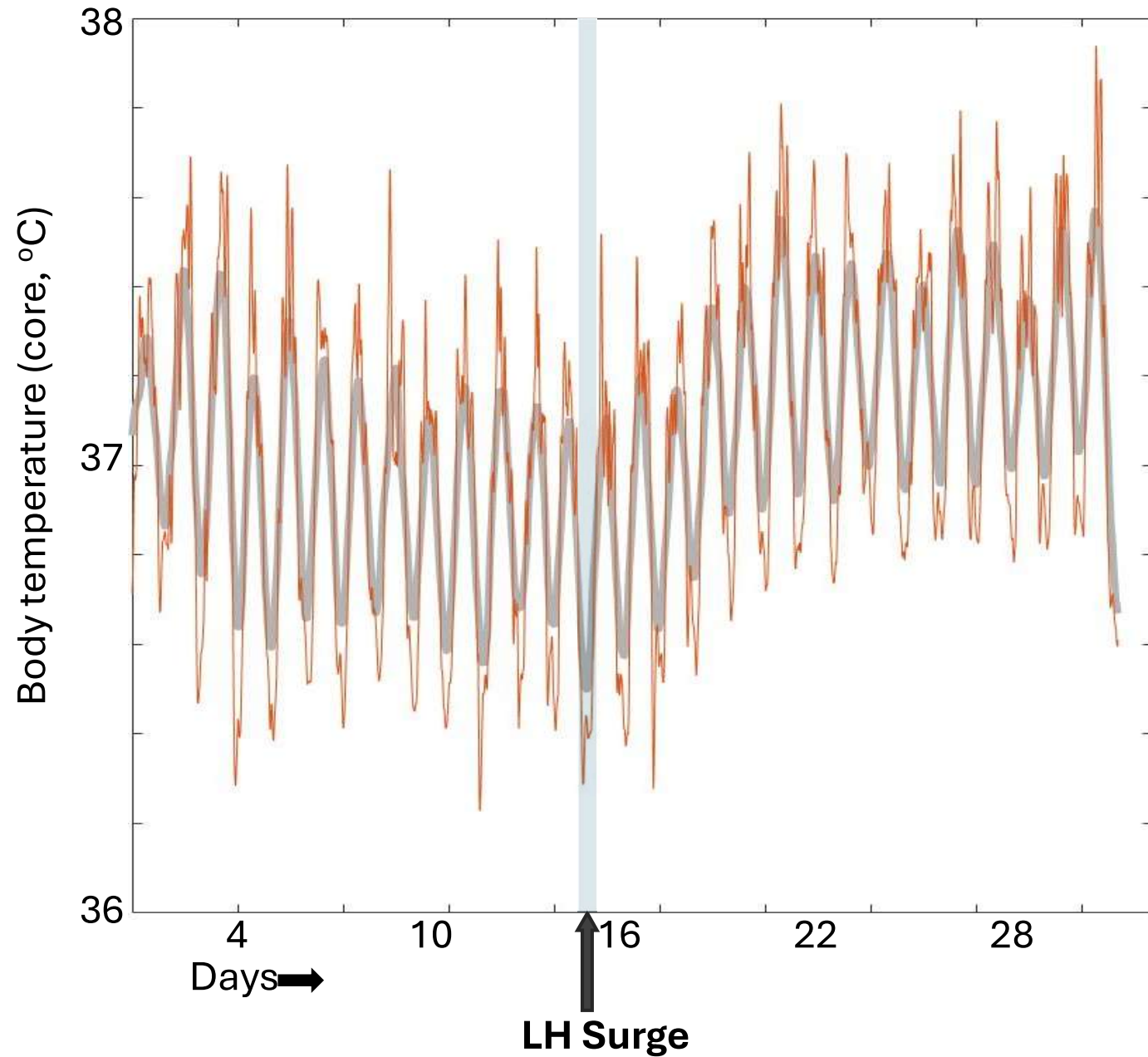
➔ Bias against blue



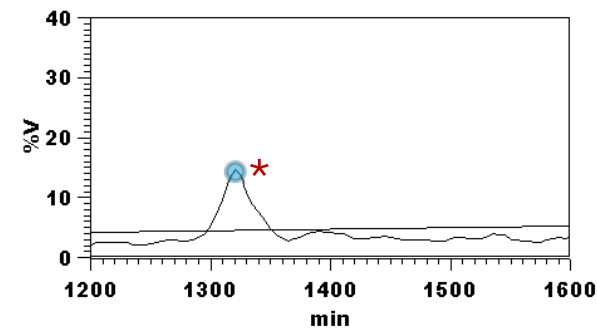
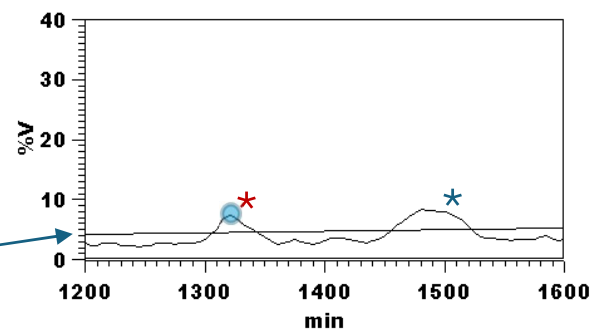
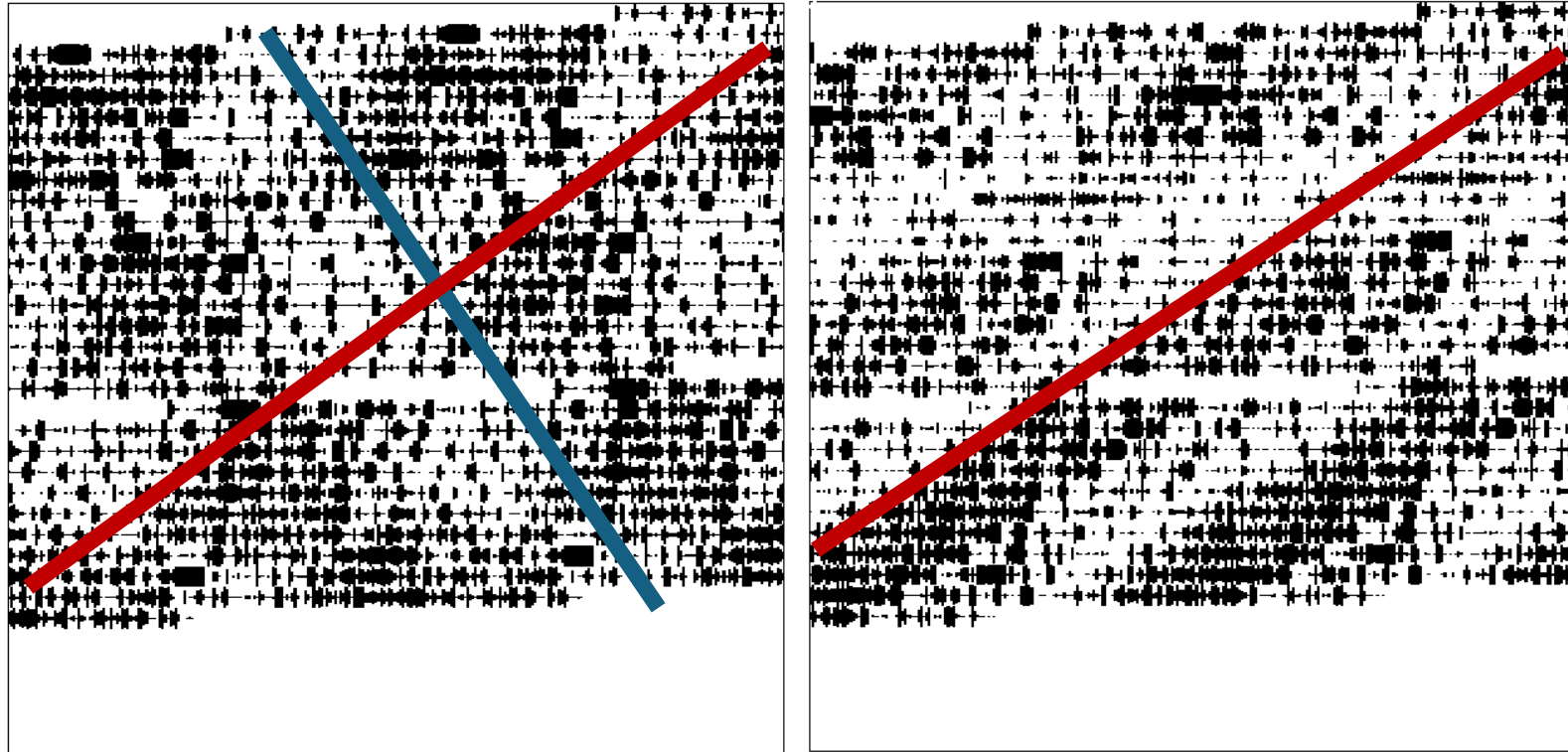
Things are always more complex than they look



Women with menstrual cycles got false alarms!



Periodograms pull out frequency & power

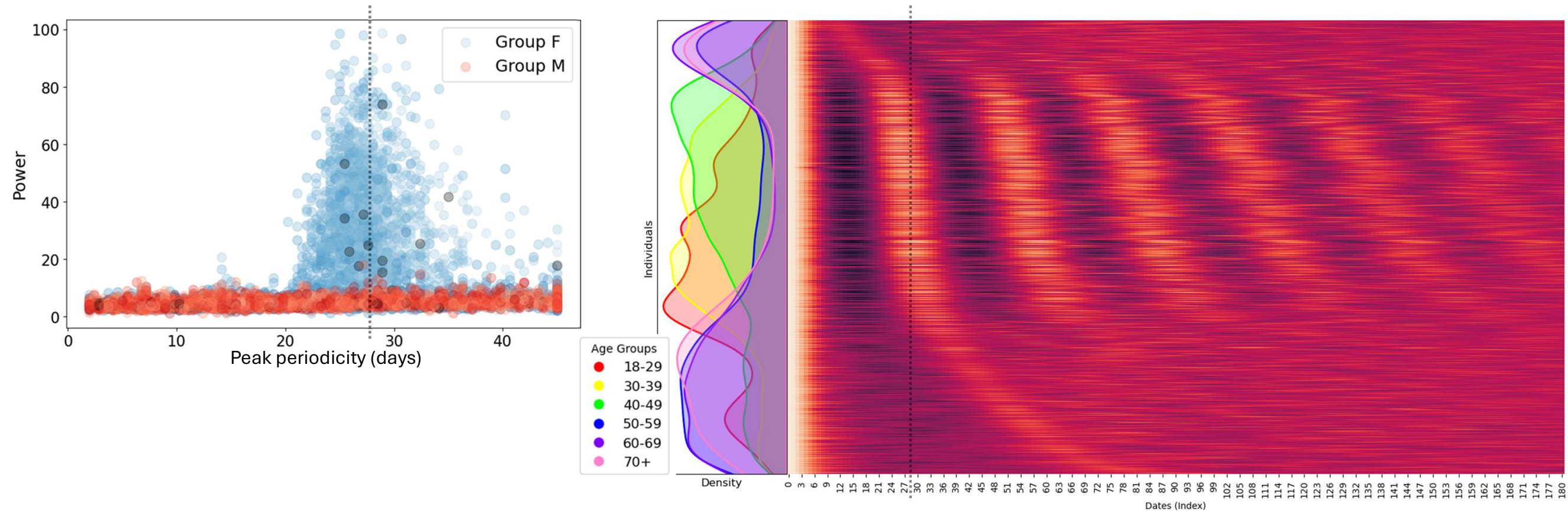


Significance line (α)

Differences we know about vs those we don't

Demographics were data-light proxies for complex systems: people

We're becoming data-rich enough to afford a demography glow-up

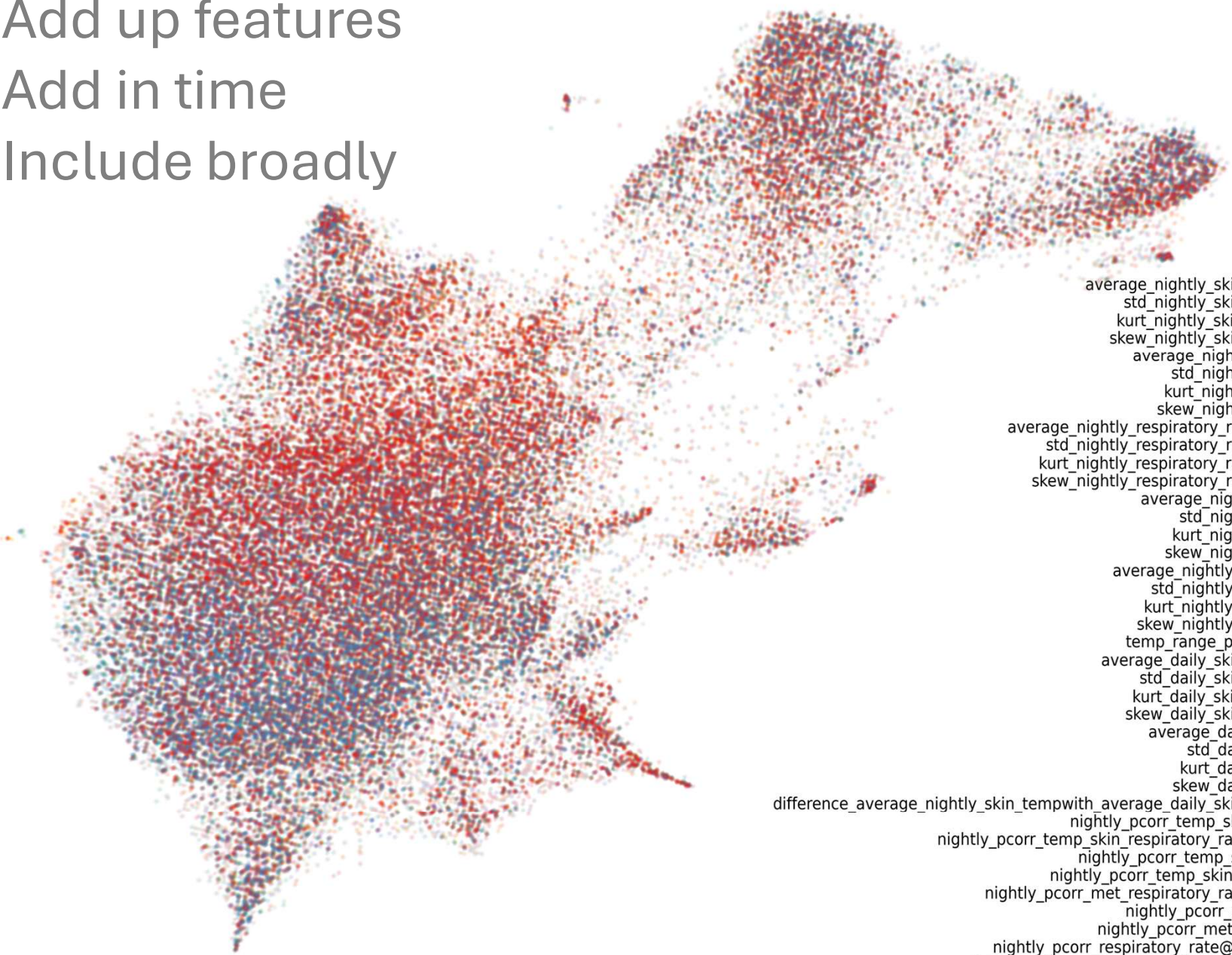


How can we explore data landscapes?

Add up features

Add in time

Include broadly



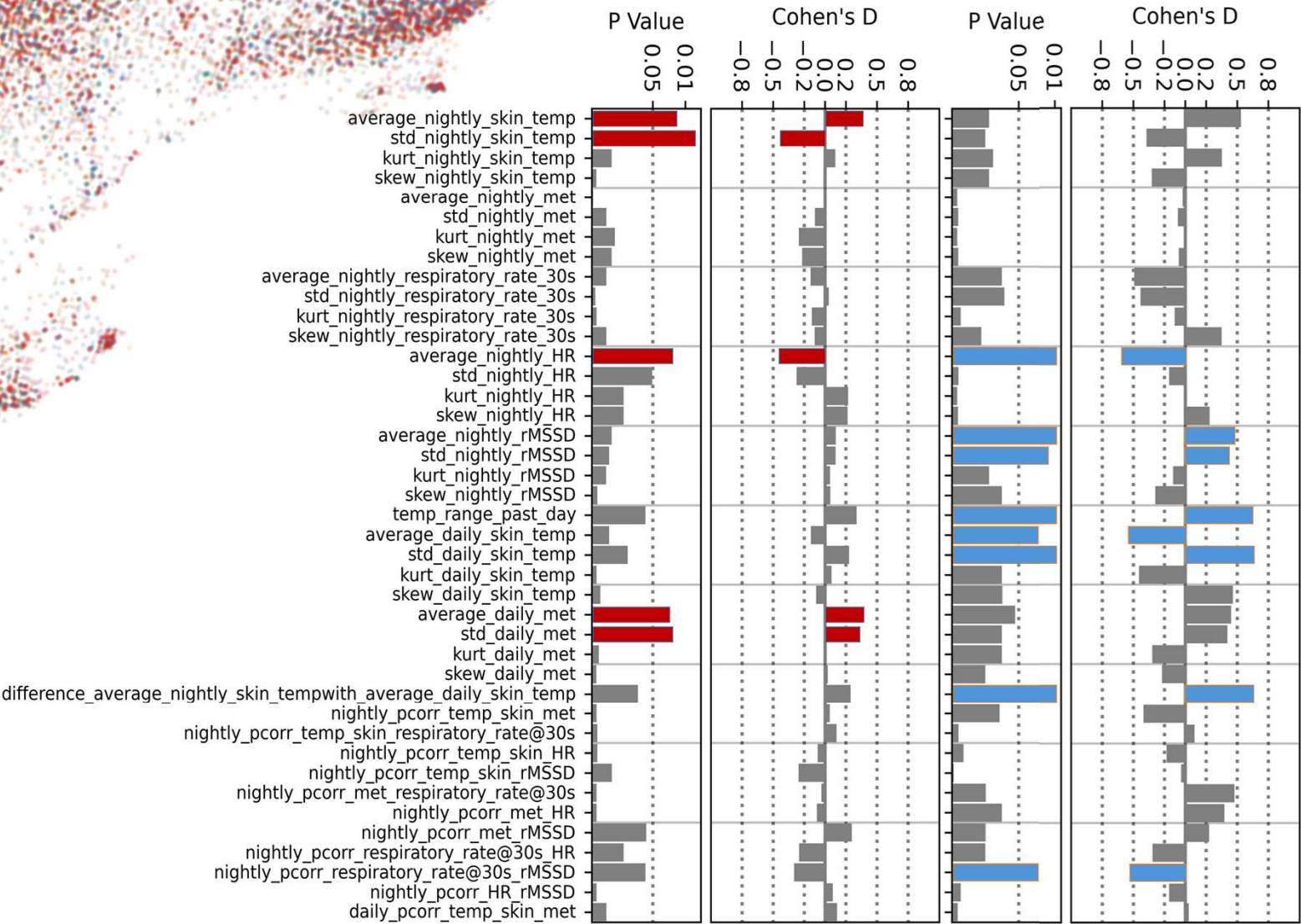
Guided approach:

-/+ Hypertension

40-50 yo

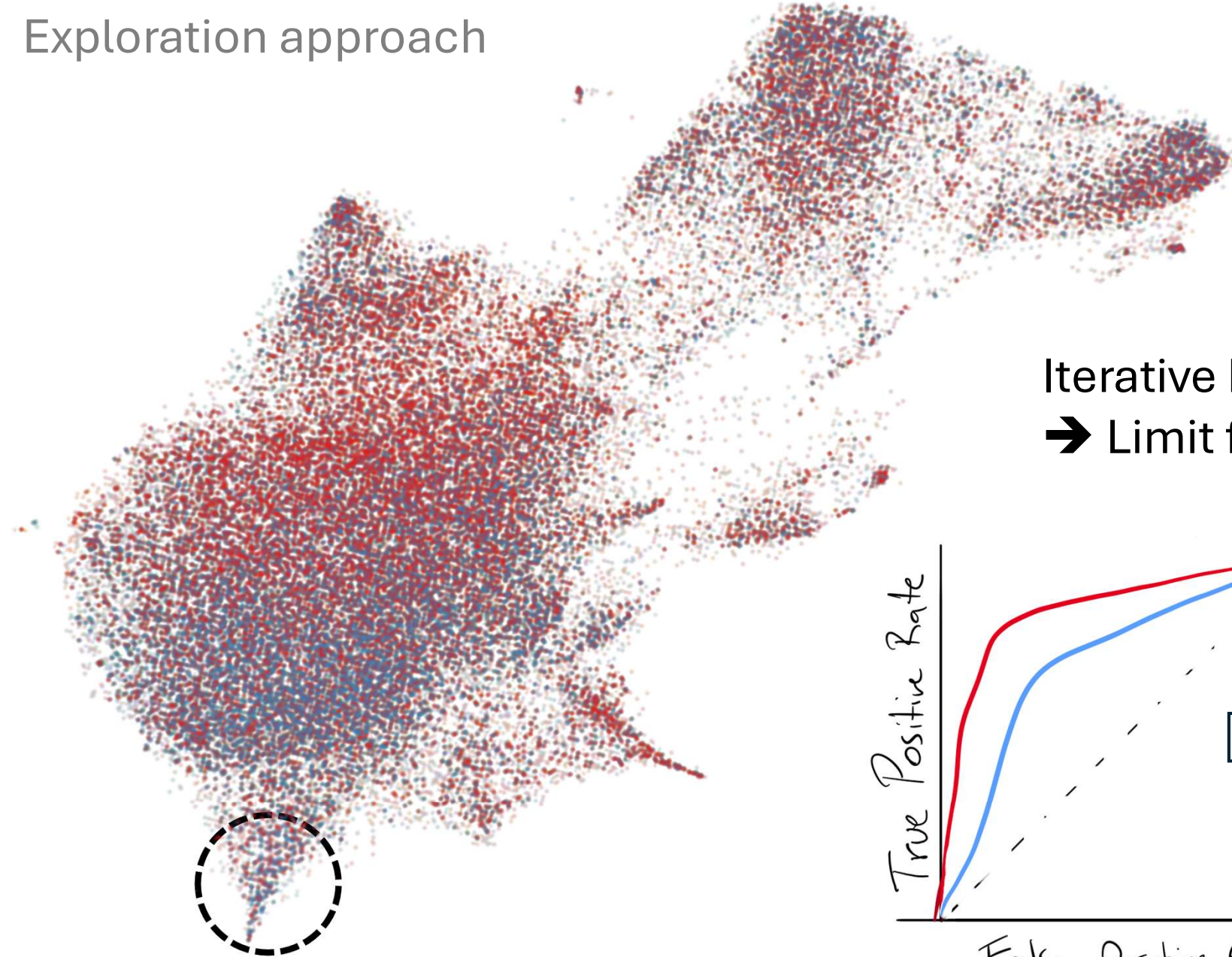
Men

Women



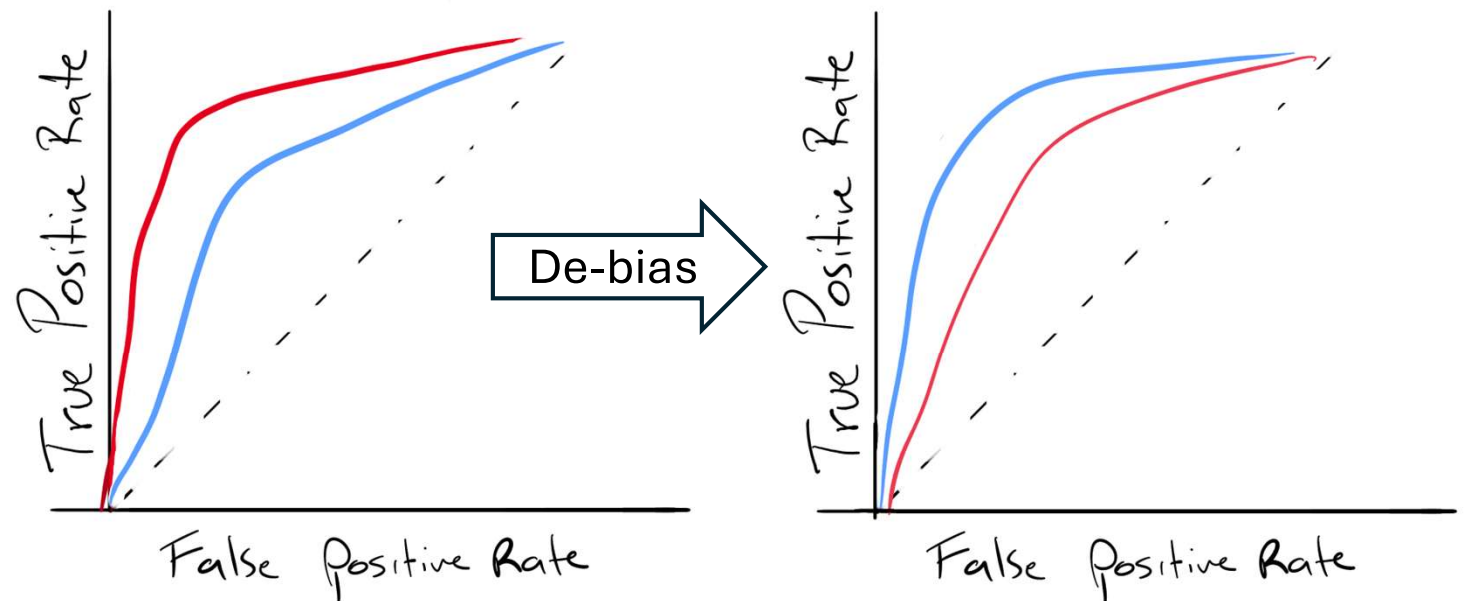
AI failures can help identify unknown groups

Exploration approach



Iterative branching and reanalysis

➔ Limit function of precision on pop



Consider: biology is more complex than language

LLMs on language prediction:

Sentences are complicated but not complex.

We know the full state of every sentence.

Sentence accuracy is 100% verifiable upon creation.

LLMs on biological prediction:

Organisms are complex but bounded by inputs.

We have very few stable ground truths.

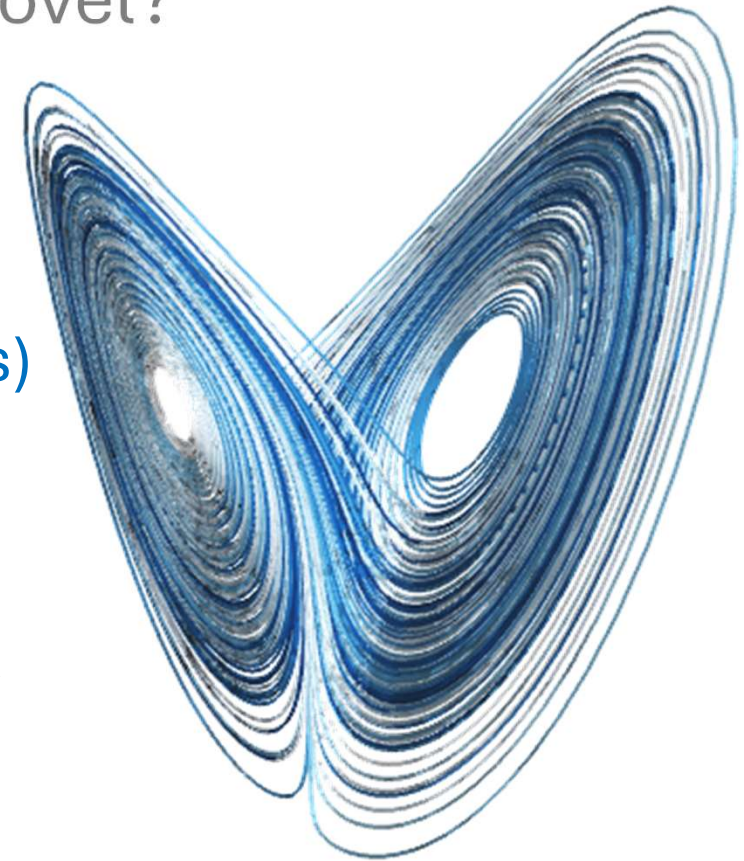
Prediction accuracy can never be fully evaluated (in many cases)

“you have COVID” *can*

“you are aging well” *cannot*

*Because of bounding, predictions on biological systems are **not random, but must be probabilistic** – recall non-random differences across people (i.e. potential for bias).*

Ben is a unique model.
LLMs can write a novel,
but can they write Ben’s
novel?



TemPredict

UCSF Health | ŌURA



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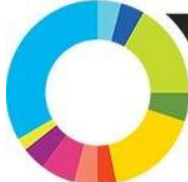
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JACOBS SCHOOL OF ENGINEERING
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 **owaves**



MTEC
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